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COMP 3008: Human-Computer Interaction
Assignment 2

1.

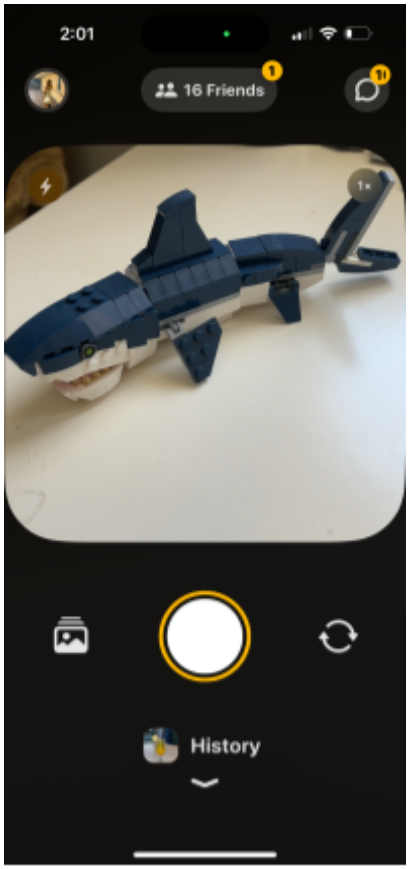

Figure 1: Locket Homescreen

a. Locket Widget App

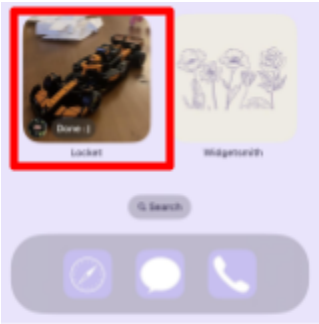

Figure 2: Widget of Locket on home screen of Iphone

One task a user might do with this interface is send a photo to a friend. The user opens the Locket widget on their home screen, a unique feature that sets it apart from standard app icons (Figure 2). Then the user taps it to activate the camera, takes a quick selfie or snaps a photo of what they're doing, adds a small emoji or short caption, and hits send. Within seconds, the photo appears directly on their friends' home screens, allowing users to share moments instantly and stay connected.

b. In the Locket Widget App, a user would engage in divided attention when simultaneously viewing a live photo update from a friend while typing a short text reaction beneath it. Viewing the photo engages visual-spatial cognition, as well as uses foveal vision for color and detail, while typing the reaction engages verbal-response. According to Multiple Resource Theory (MRT), these two activities use different codes (spatial vs. verbal) and occur across different stages (cognition vs. response), which reduces interference and allows them to be performed at the same time. To design this effectively, I would keep the photo clearly visible at the top and the text box fixed below, so users can easily switch attention between the image and typing. This setup uses MRT by keeping the tasks in separate resource areas, allowing divided attention without overloading the cognitive load of a user.

c. An example of a gulf of execution in the Locket widget app is when a user wants to add a friend. Currently, to add someone, the user must go to their profile, select "Friends," and then press the hidden "Add Friend" option. While the goal, to connect with a friend is obvious, the steps required are hidden and not immediately clear, creating a gulf of execution. To improve this, the interface could use natural mapping and perceived affordance by placing a clearly labeled "+ Add Friend" button in a prominent location on the main screen, perhaps near the friends' list. Using a familiar icon, such as a "+" next to a person's silhouette, would make the function more intuitive and align with the user's mental model of adding contacts. Most

users are familiar with the Contacts app on their phones, where the “+” button is universally used to create a new contact. By using this learned convention, users can immediately understand that tapping the button in Locket will add a friend, reducing confusion and bridging the gulf between their goal and the required actions.

After sending a photo, users may not know whether it was successfully delivered to their friend’s widget, creating a gulf of evaluation because the system state is not clearly communicated. Currently, the app shows a checkmark immediately, even if the photo is still in the process of sending, which can be misleading. To improve this, Nielsen’s design principle of visibility of system status can be used by displaying a loading or progress indicator while the photo is uploading, and only showing the checkmark once the transfer is fully complete. Additionally, applying Shneiderman’s principle of designing dialogs to yield closure, the checkmark appears only after the photo is fully sent, providing a clear signal that the action is completed. This also uses perceived affordance, as users can intuitively understand that the checkmark represents successful delivery. These design elements help users understand the system’s status and bridge the gulf of evaluation.

d. A user may tap the photo icon expecting to open the camera, but instead, the app takes them to the premium version. This is a skill-based slip, specifically a capture error, because the user’s usual habit (tapping a photo icon to open the photos) takes over, even though the app is doing something different. The user expects one action based on past experience, but the interface does not match that expectation, which can cause confusion and a capture error. To prevent this, the app could redesign the icon so that it clearly shows which button opens the camera and which button is for premium features. For example, the camera icon could look like a standard camera, while the premium option could use a different symbol or label. This uses the consistency and standards principle, because the icons would follow familiar patterns that users already know from other apps, and the perceived affordance principle, because the design would make it obvious what each button does, helping users predict the outcome of their actions and avoid mistakes.

An example of another mistake a user might make while using the app is a knowledge based mistake. For example, a user might try to take and send a photo while their device has no internet connection, expecting the photo to be delivered immediately. This is a knowledge-based mistake because the user does not fully understand how the system works in this new situation. They assume the app can send the photo without a connection, which is not possible.To prevent this, the app could provide a clear warning or message as soon as the user tries to send a photo without internet, such as “NO CONNECTION: Photo will be sent when you are back online.” This uses Wicken’s knowledge in the world principle, because the system is giving the user important information externally, instead of relying on them to remember how the app behaves. It also uses Shneidermain's feedback principle because the

warning immediately informs the user about what is happening with their photo and what the system is doing.

2.

a.

Goal 1: To assess whether users can effectively send and receive photos through the widget without errors or delays. This goal directly connects to the usability goal of effectiveness. The app is designed to let friends share photos instantly, so observing how well users can complete this task shows whether the system is performing as intended. Watching for issues such as photos not appearing on friends' widgets, delays in delivery, or uncertainty about whether a photo was sent will reveal gaps in the app's effectiveness. Understanding these problems is important because if photos cannot be shared, the app fails at its main function of sharing photos.

Goal 2: To evaluate how easily new users can learn to set up the widget, add friends, and send their first photo. This goal connects to learnability, since new users must be able to start using the app quickly without confusion. Observing how users approach setup, navigate menus, and complete their first photo share will highlight where the interface feels intuitive and where it causes hesitation or mistakes. If tasks like adding friends are hidden or unclear, users may struggle to get started, which can lead to frustration and the user may stop using the app. Learnability ensures that the app supports first time users and encourages continued use.

Goal 3: To examine whether using the app is enjoyable and creates a sense of connection between users.This goal links to the user experience aim of enjoyability. Locket is meant to strengthen social bonds by letting friends or family share small daily moments, so it is important to observe whether users react positively to receiving photos, feel closer to the sender, and engage with the app in a meaningful way. If users instead feel bored, frustrated, or disconnected, the app is not fulfilling its goal of creating enjoyable, emotionally rewarding interactions. Observing enjoyment helps determine whether the app succeeds in building a sense of happiness and connection that makes people want to keep using it.

b. The observations targeted the three usability goals from Assignment 1, effectiveness (can users reliably send and receive photos), learnability (how easily new users set up and use the widget), and enjoyability (whether the app produces positive social/affective experiences). The techniques below were chosen to help target the goals. I observed two participants using the Locket widget app in their own homes at times that were convenient for them. I chose participants' home settings to increase ecological validity as their behaviour on a familiar device and in a natural environment is more likely to reflect real use than a lab session. Home

locations also let me note environmental factors (lighting, interruptions, background noise) that could affect tasks like taking photos and therefore relate to effectiveness and enjoyability. Furthermore, I choose the most convenient location for my users and I to meet. I chose the duration to be a few minutes, as the app's core tasks are quick and keeping observations short keeps the 'Think Aloud' method speech natural. The two participants are two different genders, different ages and one has used the application before and one has not to diversify the observation.

Materials:

- **Participants' own phones:** using each person's personal device avoided complications such as learning to use a new device and allowed observation of how the app performs on real world hardware, relating back to effectiveness.
- **Think-aloud protocol:** participants were asked to verbalize their thoughts as they worked through tasks. This method was chosen because it reveals users' mental models and shows real time reactions .
- **Observer materials:**
    - Written field notes (piece of paper and pen): sed to quickly capture key observations in real time, though some subtle behaviors or details may be missed if just relying on an audio recorder.
    - An audio recorder and transcription application (App: Voice Memos): Used to capture all verbalizations accurately for later review and detailed analysis, ensuring no user comments are lost.

Procedure:

- **Consent & briefing:** I confirmed voluntary participation, and asked permission to take notes and record audio. I reminded participants they could stop anytime.
- **Think-aloud:** I gave a brief demonstration of the think-aloud method and asked them to practice once on a simple unrelated action. This reduced unnatural silence and improved the quality of verbal data.
- **Task:** Participants performed the task of taking a photo and responding to a photo while thinking aloud. I kept my role to an observer only, I did not help with the tasks or make any comments during the task. If the participant stopped talking for more than 20 seconds I used neutral prompts rather than leading questions.
- **Post observation debrief:** After the tasks were completed, I asked users if they enjoyed the observation and if they had any feedback for me.

c.

User 1 Transcript:

[Informed user to start on his home screen and open the app, send a photo and respond to a photo. Informed user "I will just be observing you and making no comments, please vocalize

all your thoughts and actions"] I'm on my home screen. I'm moving to locket *<< click>><< goes to locket widget located on his homescreen, which opens the application when clicked >>*. Not enough photos *<<error popped up in the app>>*. What is that supposed to mean? Okay. So that's not fun. *<<click>> <<pressed 'Continue' and goes to app homescreen >>* Oh, I got a picture. *<<user referring to a photo received from another user>>* I'm sending many hearts *<<click>> << goes to press emojis>>. <<swipe>> << slid up the screen and viewed the preview of the photo>>* And I'm sending a picture now. Why does it make me look like that? I'm sending the picture. *<<click>> << pressed the 'Send' button under the photo to send a photo>>* And now I'm going to respond *<<click>><<goes to the 'Messages' portion of the application>>,* but I can't reply to it. *<<user tried to directly reply to a specific individual message by sliding it (not supported by application)>>* and it's hard to send emojis. *<<click>> << goes to 'Locket Gold' page>>* Woah, well now I'm at Locket Gold somehow. What is the price? 50 bucks a year. What? That's crazy. To get.. Wait, unlimited friends. How many friends can I have? *<<click>> << goes to 'Friends' page >>* I can only have 20 friends. I don't have to pay for it. What the heck is that? Okay, what else do you get with Locket Gold? To put a frame on my widget? That's kind of awful.Yeah, that's kind of everything, I sent a photo and responded to one.

User 2 Transcript:

[Informed user to start on his home screen and open the app, send a photo and respond to a photo. Informed user "I will just be observing you and making no comments, please vocalize all your thoughts and actions"] So I see a locket photo on my widget *[homescreen]*. I'm gonna press on the widget. *<<click>> <<App opens to app homescreen>>* Oh, my locket pops up and asks me to try to track my activity across the company's apps and websites. I'm gonna say no *<<click>> <<user clicks no and proceeds to app>>* then I'm going to take a photo. I'm a little confused on what that's *[the user is looking at the home screen where she can take a photo]* gonna do. And then I guess it tells me to send it to someone and out a message. *<<click>> <<took a photo>>* I'm going to take a different photo. *<<clicks>> <<pressed the "x" button to get rid of the photo>>* I'm gonna send that off. Oh, I pressed the check mark. *<<click>> <<photo was sent out>>* Okay, that was confusing. I think I sent it without meaning to. Okay, so I guess I've sent the photo. I would say that's not very usable, because I saw there were options for a caption. I didn't get to do that. I just pressed the checkmark and it was gone. Then I scroll and I can see what other people send. *<<slid up>> <<proceeds to see photos sent from friends>>* Ooh, I can send a message. I sent a fire emoji to someone *<<click>> <<pressed on emoji to send>>* , but I don't know how to respond to them.Can I? Okay, this is another photo. Oh, I can send a message to someone specifically. I'll send that. *<<click>>* Send. Okay, now I see there's messages *<<click>> <<goes to the "Messages" portion of the app>>* , a message aside. Okay, I see those messages. I'm not interested in that. *<<click>> <<goes back to app home screen>>* Oh, I see there's a friend's option. *<<click>> <<goes to the 'friends' portion of the app>>* Ooh, I have friend requests.

I'm not sure who these people are. Why do I need 20 friends? Okay, I see I see it's from my contacts. *<<clicks>> <<goes to 'history'>>* then I can see my history and I can see what everybody else posted. How do I delete one of my photos? I want to try to delete something. I don't think I can.. Oh, that's 'cause it's somebody else's. I want to try to delete this? *<<click and hold>>* Is it there forever? We'll click it. *<<click>> <<goes to the 'Everyone's History' portion of the app>>* Oh. Okay, I want to delete what I posted, but I can't see how to. I'm pressing on the photo and nothing's happening. It'll be here. I'm seeing all the photos. When I press the four buttons *<<click>>*, I see what everybody's posted, but I still don't see how to delete mine. Ooh, I found it. When I press share *<<click>>*, I can delete. I don't think that's very readable. But that's okay. *<<click>> <<goes to 'My Profile>>* Then I go on my profile. Oh, and I can see all that I posted. and I can see. I don't know what this is *[A pop up for 'Rollcall']*. Ooh, roll call, a weekly photo dump with your best friends. I'm not interested. And then I see the settings.And, yeah, I think I'm done.

d.

User 1 Observation Notes	
Activities:	Tasks - Open application through widget - Respond to a received photo - Send a photo Steps taken - Clicked widget → app opened → encountered error message ("Not enough photos") - Sent a photo and added emoji reactions - Tried to reply directly to a message by sliding it (unsupported feature) - Accidentally navigated to Locket Gold and reviewed subscription details Errors - Confusion at unclear error message ("Not enough photos") - Frustration at inability to reply to specific messages - Mis-tap led to accidental navigation to paid premium page
Environments:	- Basement couch, sitting beside each other at user's house - Private, comfortable space - Minimal distractions, but user was distracted by app interruptions (error pop-up, redirect to premium features) - Home setting supports naturalistic use, but system design issues disrupted flow more than environment did

Interactions:	- With the application:
	  - Engaged with app interface (widget, buttons, emojis, messages, premium page)
	  - Attempted natural gestures (swiping to reply) that weren't supported
	
	- With others:
	  - Indirect interaction with friends via photo sending and receiving
	  - No direct human interaction during observation (besides researcher presence)

Objects:	- Primary device: User's own smartphone (Iphone 12, App version: 2.14.1) - Physical objects: None beyond the phone; the environment was simple

Users:	- Role: Participant acting as a casual Locket app user
	- Frustrations:
	
	  - Error message not clear
	  - Limited interaction options in messaging
	  - Unwanted navigation to premium features
	  - Perceived premium cost as excessive
	
	- Preferences & expectations:
	  - Expected to reply to messages directly via swipe (based on prior messaging app experiences)
	  - Preferred clarity in app communication (error messages, premium features)
	
	- Limitations:
	  - Cognitive reliance on prior habits (swipe-to-reply from other apps)
	  - Unfamiliarity with premium structure caused confusion

User Information:	- Male - Age 20 - Uses Locket frequently (a few times a week)

User 2 Observation Notes	
Activities:	Tasks - Open application through widget

	- Send a photo - Respond to a received photo Steps taken - Opened widget → App asked about tracking → Chose "No" - Took photo → Deleted → Sent new photo (confusion about checkmark sending automatically) - Scrolled through others' photos → Sent emoji reaction → Tried to reply to photos → Opened "Messages" tab Explored "Friends" and "History" → Tried to delete a photo (unclear process) → Found delete under "Share" option - Checked "Profile" and "Roll Call." Errors - Confused by the checkmark auto-sending photo before adding caption. - Struggled to delete a photo (unclear steps, deletion option hidden under "Share"). - Confused about the "20 friends" limit. - Didn't understand the "Roll Call" feature at first glance.
Environments:	- Living room couch, sitting beside each other at user's house - Private, comfortable space - A few noises from user's family (in the kitchen making noises, and loud chatting) - Disruptions from within the app: pop-ups, hidden features, unclear actions - Home setting supports naturalistic use

Interactions:	- With the application:
	  - Engaged with app interface (widget, buttons, emojis, messages, friends list, history, profile)
	  - Attempted gestures like pressing, holding, and tapping icons (expected certain actions but not always supported).
	
	- With others:
	  - Indirect interaction with friends via photo sending and receiving
	  - No direct human interaction during observation (besides researcher presence)
	  - Noises from user's family members were heard in a different room

Objects:	- Primary device: User's own smartphone (Iphone 14, App version: 2.14.1) - Physical objects: None beyond the phone; the environment was simple

Users:	- Role: First time Locket app user.
	
	- Frustrations:
	  - User felt confused when photos were sent too quickly (lost opportunity to add caption).
	  - Difficulty finding delete option for photos.
	  - Unclear on friend requests and app limitations (20 friends).
	  - Pop-ups like Roll Call distracted from the main task.
	
	- Preferences/Expectations:
	  - Expected captions to be added before confirming photo send.
	  - Expected easier, more visible way to delete own photos.
	  - clarity in interface labels and actions were not clear
	
	- Limitations:
	  - Relied on prior experience with other apps (e.g: simple delete functions).
	  - Cognitive load increased due to hidden functions and non-intuitive steps.
	  - Incomplete understanding of new features (Roll Call, friend limits).

User Information:	- Female - Age 25 - Has not used Locket before

e.

A key insight from observing User 1 was that the app often failed to support effectiveness, which directly aligns with Goal 1: assessing whether users can send and receive photos without errors or delays. For example, when the error message "Not enough photos" appeared (User 1 Transcript), he said, *"What is that supposed to mean? Okay. So that's not fun"*, indicating confusion and uncertainty about what action to take. This lack of clear feedback reduced his confidence that the app could reliably send and receive photos, undermining its core purpose of photo sharing. Furthermore, when he attempted to reply to a specific message by swiping, a gesture familiar from other messaging apps, he was unable

to do so, reflecting a violation of the consistency and standards principle. These observations demonstrate breakdowns in effectiveness and error prevention, as hesitation and unsupported actions prevented smooth task completion. Applying user-centered design principles, such as naturalness (matching design to mental models of messaging apps) and perceived affordances (e.g., swiping should afford replying), could help the app better support effective interactions (Observation Notes, User 1).

From User 2's experience, challenges were more closely tied to learnability, which supports Goal 2: evaluating how easily new users can start using the app. As a first time user, she struggled with hidden features. For instance, the checkmark that autosent her photo before she could add a caption, stating, *"Okay, that was confusing. I think I sent it without meaning to"* (User 2 Transcript). She also found the delete function buried under the "Share" option, showing a breakdown in the user control and freedom principle because the app did not allow her to complete tasks in the expected way. Her confusion over friend limits and the "Roll Call" feature further increased cognitive load, requiring extra mental effort to understand the interface (Observation Notes, User 2). These usability barriers also affected Goal 3 on enjoyability: both users expressed frustration with unclear processes (User 1 regarding premium costs, User 2 regarding deletion), which risked reducing engagement and positive emotional experiences. Leveraging design principles such as Gestalt's proximity principle (e.g., grouping captions and send buttons clearly) and Shneiderman's feedback principle (providing confirmation and undo options) could enhance clarity, reduce cognitive load, and improve overall enjoyment, supporting the app's goal of creating meaningful connections through photo sharing.

3.

a. This was essentially a pilot study; what would you have done differently in your observations if you were doing a full study in real life?

If this were a full study rather than a pilot, I would expand both the number of participants and the diversity of contexts to get a more complete understanding of user behavior. In a real life study, I would recruit users with different levels of familiarity with the Locket app, different age groups, and devices, to see how these factors influence effectiveness, learnability, and enjoyability. I would also conduct the observations in a wider range of real world environments, such as while commuting, at school, or in the middle of a conversation, to better capture naturalistic usage. This would allow me to see how context affects interactions, errors, and emotional responses, rather than only relying on the controlled and comfortable home setting used in the pilot.

Additionally, I would incorporate guided reflection questions at multiple points during the session rather than only relying on think-aloud commentary. After users complete key tasks

like sending a photo, responding to a message, or managing friends, I would ask them how they felt about the task, whether it was frustrating or enjoyable, and whether they understood what was happening. This would provide richer qualitative data on emotional reactions and satisfaction, which directly ties to Goal 3: enjoyability. Capturing users' reflections immediately after actions would reveal subtle frustrations or delights that might not be obvious from observation alone, giving a more complete understanding of the app's user experience.

b. Did your observation results match your expectations? Was there anything surprising?

The results of the observation generally matched my expectations in that both users encountered difficulties with hidden or nonintuitive features, which directly aligned with the usability goals of effectiveness and learnability. These findings confirmed my hypothesis that even experienced users could face challenges when navigating unclear interface feedback, such as error messages, or when attempting to use features that are not immediately obvious. Observing these challenges reinforced the importance of applying design principles like visibility of system status, user control and freedom, and perceived affordances to help users understand the system and complete tasks without confusion. The pilot study highlighted how critical it is for Locket to communicate clearly, guide users through actions, and ensure that the interface supports both first time and frequent users in achieving their goals smoothly.

What was particularly surprising, however, was the degree to which users' prior experiences with other apps shaped their expectations and interactions with Locket. Both users instinctively attempted gestures or actions that are common in other messaging and social media apps, but these actions were not supported in Locket, resulting in confusion and discontent. Additionally, peripheral features such as premium subscriptions, friend limits, and the Roll Call function created unexpected distractions, leading to confusion and missteps. These observations emphasized that usability testing needs to consider not just the core photo sharing tasks, but also secondary features that could disrupt the user experience. In a full study, I would also ask users direct questions about their emotions, satisfaction, and overall enjoyment, allowing me to better assess Goal 3: enjoyability and gain a deeper understanding of how engaging and meaningful the app is in promoting social connection.